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EXPT.NO - 7a

TITLE - 0/1 Knapsack problem using Dynamic Programming

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In [2]: 1 # Taking inputs from user
2 n = int(input("Enter the number of items: "))
3
4 print("Enter the weight and profit for each item:")
5 weight = []
6 profit = []
7
8 for i in range(n):
9     w = int(input("Weight of item {}: ".format(i+1)))
10    p = int(input("Profit of item {}: ".format(i+1)))
11    weight.append(w)
12    profit.append(p)
13
14 capacity = int(input("Enter the capacity of the Knapsack: "))
15
16 def DP_KnapSack(capacity, weight, profit, n):
17     # Create DP table
18     K = [[0 for x in range(capacity + 1)] for x in range(n + 1)]
19
20     # Build table K[][] in bottom-up manner
21     for i in range(n + 1):
22         for w in range(capacity + 1):
23             if i == 0 or w == 0:
24                 K[i][w] = 0
25             elif weight[i-1] <= w:
26                 K[i][w] = max(profit[i-1] + K[i-1][w-weight[i-1]], K[i-1][w])
27             else:
28                 K[i][w] = K[i-1][w]
29
30     return K[n][capacity]
31
32 max_profit = DP_KnapSack(capacity, weight, profit, n)
33
34 print("=====")
35 print("Maximum Profit Earned = ", max_profit)
36 print("=====")
```

```
Enter the number of items: 3
Enter the weight and profit for each item:
Weight of item 1: 10
Profit of item 1: 60
Weight of item 2: 20
Profit of item 2: 100
Weight of item 3: 30
Profit of item 3: 120
Enter the capacity of the Knapsack: 40
=====
Maximum Profit Earned = 180
=====
```

